RIVER VALLEY HIGH SCHOOL
JC 2 PRELIMINARY EXAMINATION

CANDIDATE NAME	

CENTRE NUMBER	S					INDEX NUMBER					

H3 BIOLOGY **9816/01**

Paper 1 **24 September 2021**

2 hours 30 minutes

Candidates answer on the Question Paper.
Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the space provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiners' Use	
Section A	
1	
2	
Section B	
3 / 4	
Total	

This document consists of **17** printed pages and **3** blank pages.

Section A

Answer **all** the questions in this section.

1	You should read through the whole of the insert and **Question 1** carefully and then use the information provided to answer the questions. The insert includes Table 1.1, Fig. 1.1, Fig. 1.2, Fig. 1.3 and Fig. 1.4. The entire question is based on information about evolution of multicellularity in animals.		
	(a)	Discuss the extent to which the endosymbiotic origin of eukaryotes conforms to cell theory.	[2]
		Conform 1. mitochondrion and chloroplast which has prokaryotic origins a. are able to reproduce independently from pre-existing mitochondrion and chloroplast via binary fission;; OR b. contains hereditary material in the form circular DNA to pass down to daughter mitochondria/chloroplast;; Does not conform 2. mitochondrion and chloroplast outside of a eukaryotic cell are unable to reproduce independently from existing mitochondrion;;	AO A
	(b)	Using your knowledge on the central dogma and the findings on the mode of genetic transfer in the insert, suggest how genetic transfer of mtDNA took place from the mitochondria to the nucleus.	[3]
		1. Transcription of mtDNA genes by (mitochondrial) RNA polymerase to form pre-mRNAs which exit mitochondria and enter nucleus via pores;; 2. Pre-mRNAs undergo post-transcriptional modification / splicing to form mature mRNAs;; 3. Reverse transcription of mature mRNAs to form double-stranded complementary DNAs which integrate into nuclear genome (via non-homologous recombination);; Accept other plausible mechanism	AO B
	(c)	The mtDNA of extant species includes at least one of the genes coding for rRNA. Explain the reason for retaining at least one rRNA gene in the mtDNA.	[2]
		1. Gene products of rRNA and ribosomal proteins need to be in the same location / co-localisation for assembly into functional ribosomes;; 2. Ribosomes too large to pass through membrane proteins if assembled outside and therefore cannot enter mitochondria;;	AO B

	(d)	**(i)**	Using Fig. 1.1 and Fig 1.2, identify three features and the related molecules that must have evolved **and** explain the significance of **each feature** in the formation of multicellular organisms.	[3]
			(see table below)	AO B
		(ii)	Unicellular organisms exist even after multicellular organisms evolved. Describe one selective advantage of each type of organism.	[2]
			(see answer below)	AO A

feature	molecules	significance in the formation of multicellular organisms
gene regulation	activators/repressors/specific transcription factors	allows cells to differentiate / specialise for division of labour, increasing reproductive fitness;;
cell-cell adhesion	glycoproteins	allows individual cells to attach to form structures made of many cells;;
cell-cell communication	receptors and ligands	allows individual cells to communicate and coordinate cellular processes / responses due to environmental triggers;;

Unicellular

1. Short reproduction duration / reproduce quickly in suitable environments so as to outcompete variants / other species;;
2. Lower energy and resource requirements needed for survival, therefore able to survive even in low resource environments;;
3. Single genome per organism therefore more significant impact of mutations giving rise to higher variation in species and greater potential for survival in changing environments;;

Multicellular

4. Able to have somatic and germline cell lines for genetic stability in future generations;;
5. Able to thrive in different environments / changes in environmental factors (temp./humidity/nutrients type & amt.) due to homeostatic ability therefore increasing the habitat range;;
6. Gain greater mobility to escape predator, hunt prey / move towards nutrient sources;;
7. Able to live longer as organism remains functional even when individual cells die, therefore increasing ability to survive / fitness till reproductive age;;

One from each type of organism

	(e)	Using your knowledge of stem cells and Table 1.1, Fig 1.3 and Fig. 1.4, discuss how genetic and molecular differences between LUA and LCMA resulted in the differences observed in the multicellular stage.	[5]
		1. More genes / gene variants compared to LUA, QV;; 2. Allows expression of different sets of genes resulting in different sets of gene products in groups of cells;; 3. Spatial distribution of molecules in LCMA, which is absent in LUA, QV - Ref. to range of concentrations of relay / signalling molecules along the axis;; 4. Concentration-dependent activation of downstream proteins in different regions of the multicellular stage giving rise to different germ layers (endoderm, mesoderm, ectoderm);; 5. Temporal distribution of molecules in LCMA, which is absent in LUA, QV – Ref. to changes in concentrations of molecules at different hours after formation of multicellular stage;; 6. Time-dependent expression of molecules allows for sequenced development/differentiation of cells / segregated differentiation of cells within a multicellular mass;;	AO B
	(f)	Compare the regulation of enzymes in a cell and the regulation of cells in a multicellular living organism.	[4]
		Similarities 1. Compartmentalisation of enzymes and cells into localised areas e.g. into a subcellular region bound by membrane and forming tissues / organs;; 2. Use of feedback regulation to control degree of response e.g. feedback inhibition where end-product binds to enzyme or receptor to inhibit further response or signal cascade in cell signaling;; 3. Regulatory molecules may be chemical and / or electrical in nature e.g. ATP, ions;; 4. Regulation of key enzymes and cell / cell types can control biochemical pathway and systemic response;; 5. Regulation of number e.g. by regulating gene expression or targeted proteolysis and by programmed cell death;; Differences	AO B

Features	Enzymes	Cells
6. Scale / speed of response	Smaller scale / more immediate effect of regulation	Larger scale / longer time taken to respond due to multi-step process;;

		7. Target of regulatory molecules	Enzymes' active site, allosteric site	Receptors on cell surface membrane or in cytoplasm of cell;;
		8. AVP		

	(g)	Studying how mutations in mtDNA accumulate with age is critical because of their association with numerous genetic diseases. Using the information provided in the insert and Fig. 1.5, suggest and explain **two** ways in which knowledge of mtDNA mutations could be used combat genetic diseases in humans.	[4]
		1. To increase/focus research on D-loop mutation and repair mechanisms in aging-related processes;; 2. Significantly higher mutation frequency in D-loop of mtDNA in brain and muscle cells in mothers (older) compared to pups (younger);; 3. To increase awareness to promote earlier reproduction in females;; 4. Significantly higher mutation frequency in mtDNA of all oocytes in mothers (older) compared to pups (younger);; 5. AVP;; 6. AVP;;	
			[Total: 25]

2	Recent studies show that epigenetic diversity is involved in adaptation and survival of species in changing environments. Compare epigenetic changes with mutations and discuss how epigenetic changes complement genetic diversity to contribute to long-term survival of species.	[25]

Compare epigenetic changes with mutations *[8m]*

Similarities

1. Both are changes to the genome that can be inherited by daughter cells / offspring;;
2. Both may result in phenotypic and/or physiological changes in the organism such as diseases/disorders;;
3. Both can be affected by / result due to external environmental factors;;
4. Both can show varying patterns among individuals, different tissues within an individual and in different cells;;

Differences

feature	epigenetic change	mutations
5. effect on genome	does not involve a change in DNA sequence / addition or removal of chemical groups on DNA molecule	involves a change in DNA sequence and / or DNA structure or number;;
6. effect on gene product	only affects expression of genes	may affect expression of gene and / or alter sequence and structure of gene product;;
7. timing and location of affected sites	non-spontaneous and targeted /specific sites in genome of selected cells	spontaneous and random sites in genome of random cells;;
8. source of change	due to effect of enzymes such as HAT, HDAC, DNA methyltransferase	due chemical mutagens, ionising radiation, infectious agents, DNA replication errors;;
9. function in living organisms	role in determining cell fate during differentiation process	role in generating genetic variation e.g. somatic hypermutation for affinity maturation;; To consider if generating new alleles as raw material for

		evolution (natural selection) is a normal function
10. evolutionary outcome	does not result in novel functions	may result in novel functions / characteristics that are selectively disadvantageous / advantageous / neutral;;

How Epigenetic changes complement Genetic diversity for Long-term survival

Definitions

11. Epigenetic changes include DNA methylation, histone modifications and small RNA regulation (which generates epigenetic diversity);;
12. Genetic diversity refers to the heritable variation present in gene pool / types of alleles present in species;;
    Long-term survival refers to species continuation over long periods of time (and including prokarya and eukarya such as animals, plants and fungi);;

How genetic diversity promote long-term survival of species:

13. Mutations is the only source of genetic diversity in a species;;
14. Genetic variation is further generated in a population by crossing over and independent assortment during meiosis, random mating, random fertilisation and migration between populations with different gene frequencies;;

Advantages of genetic diversity

15. Advantage 1 – Heterozygote advantage. Many disorders are recessive, therefore being heterozygote could promote increased survival rate;;
16. Ref. one example: Sickle cell anemia in malaria-prone areas. Hb^AHb^S individuals have less severe disease / sickle cell trait and better able to survive Plasmodium infections;;

A: Reverse argument Problems due to high homozygosity (percentage of identical alleles at particular gene locus): ref. one example Example: Many disorders / deleterious alleles are recessive. High homozygosity / low genetic variation results in increased likelihood of inheriting recessive alleles resulting in disorder and death;;

17. Advantage 2 – Increased survival fitness of species in changing environments;;
18. Ref. one example: Antibiotic-resistant bacteria strains such as penicillin-resistant strains of *Staphylococcus aureus* documented within same year of widespread usage of penicillin;;

19. Advantage 3 – Able to occupy wider range (geographical / biophysical) of habitats therefore increasing survival probability of species as a whole;;
20. Ref. one example: Plains spadefoot toads (*Spea bombifrons*) occupy wide range throughout Southwestern and Central United States ranging from grassland to desert habitats;;

Limitations of genetic diversity AND complementation by epigenetics:

21. [Limitation] Unable to respond directly to environment during individuals' lifetime resulting in insufficient / inadequate level of phenotypic expression;;
22. [Limitation] Inability to respond to rapid changes in environmental conditions;;
23. [Complement] Environmental factors such as diet, lifestyle, biotic factors lead to epigenetic modifications resulting in changes to gene expression;;
24. Through different expression levels to result in different phenotypes i.e. allows fine-tuning / greater phenotypic variation even with same set of alleles and genes / phenotypic plasticity;;
25. [Limitation] Low rates of genetic mutation in certain species due to molecular mechanism of replication and repair, generation time and number of offspring;;
26. [Limitation] Only genetically variable loci are capable of mutations OR genes that have essential roles and without duplicates / redundancy are incapable of / have limited capacity for mutations without causing death of individuals in the species;;
27. [Limitation] Species that undergoes asexual reproduction have limited mechanisms / only mutations that generate genetic variation;;
28. [Complement] Rate of epigenetic changes are substantially higher, therefore creating heritable variation that can enable adaptation and survival before genetic adaptation evolves;;
29. Example:
    - Trans-generational immune priming where transfer of parental immunological experience enhances offspring immune defence in both vertebrates and invertebrates;;
30. [Complement] Allows species to occupy new / unoccupied ecological niche before genetic adaptation;;
31. Example:
    - Success of invasive species such as house sparrows (*Passer domesticus*) or Japanese knotweed (*Fallopia japonica*);;
32. AVP;;

Section B

Answer **one** question in this section.

3 The most important biomolecule of a cell is the phospholipid. Discuss. [25]

Introduction (Max 5)

1. A cell is the smallest unit of life and is the fundamental unit of structure and function in living organism;;

2. Biomolecules are organic molecules including carbohydrates, proteins, lipids and nucleic acid and are important for the survival of living cells;;

3. Phospholipid structure: Phospholipids are a class of lipids that contain a phosphate group, glycerol and 2 fatty acids;;
4. Fatty acids in phospholipids may contain saturated or unsaturated fatty acids;;
5. Phospholipid property: The phosphate group of the phosphatidic acid forms the hydrophilic (i.e. water-loving) "head" of the molecule whilst the hydrocarbon chains of the fatty acid moieties form the hydrophobic (i.e. water-hating) "tails" of the molecule. Such molecules are described as being amphipathic.
6. Phospholipid function: The internal environment of the cell (i.e. cytoplasm) and its external environment are both aqueous. In an aqueous environment, phospholipid molecules assemble themselves into a <u>phospholipid bilayer</u> and is an essential component of biological membranes.

Argument for statement: Phospholipid is the most important component of living cells (max 5, max 2 without compulsory points)

7. Phospholipids form of lipid bilayers are essential components of biological membranes, all cells are enclosed by a cell surface membrane/plasma membrane which acts as a boundary between internal and external environment;; *compulsory point
8. Phospholipid forms a hydrophobic core, making the membrane selectively permeable, preventing the entry of large and polar substances into cell, allowing only small non-polar substances to diffuse into cell;;
9. Important for gaseous exchange, entry of oxygen and removal of carbon dioxide from cell for cellular respiration/ entry of carbon dioxide and removal of oxygen for cellular photosynthesis;;
10. Prevents entry of polar and charged substances, maintaining cell water potential, preventing lysis of cell;;
11. Prevent the exit of substances for use within a cell, increasing concentration of substance for cellular reaction/maintaining concentration gradient,
12. Such like the accumulation of ions to maintain resting potential of neurons/proton gradient for chemiosmosis;;
13. In eukaryotes, phospholipids form membranes of organelles allowing for presence of membrane bound organelles like nucleus/mitochondria/chloroplast/golgi apparatus/ER/lysosome;;*compulsory point
14. Quote one advantage of compartmentalisation;;
15. In eukaryotes, the fluid nature of the phospholipid plays important role in the endomembrane system of the cell;;

16. Phospholipids form membrane with lipid bilayer, which allow the embedding of proteins, steroids, glycolipid for various function of cellular transport, cell signalling and cell-cell recognition/communication;;

17. Fluid nature of phosopholipid allows for transport

Accept AVP

AVP with quoted example

Argument against statement: other lipids are also important component of living cells (Max 3)

18. Triglycerides are important as long term energy store of the cell, important for cells of hibernating animals/ valid examples;;
19. Glycolipid important for cell-cell recognition and communication and are used as cell-surface makers;;
20. Steroids are important to prevent close packing of phospholipids leading to increased fluidity of membrane so that membranes can carry out their various function

Accept AVP

AVP with quoted example

Argument against statement: nucleic acids are the most important component of living cells (Max 3)

21. Nucleic acid type and function - Nucleic acid include DNA, mRNA, rRNA and tRNA and are the most important component because they are unit of heredity/stores genetic information/template for making other components in cell;;
22. All living organisms (from prokaryotes and eukaryotes) have DNA and RNA;;
23. DNA stores information and is replicated faithfully to be pass down from one generation to the next
24. RNA including mRNA, tRNA and rRNA allow for protein synthesis, to produce enzymes and proteins to form cellular components including enzymes to make phospholipids;;
25. RNA can also catalyse reactions, e.g. ribozyme, which is used in translation.

Accept AVP

AVP with quoted example

Argument against statement: proteins are the most important component of living cells (max 3)

26. Enzymes are required biological catalyse and they catalyse a wide variety of reactions required in cells.
27. E.g. (quote 1) kinases in cell signalling allow for signal transduction/enzymes in glycolysis/Krebs cycle allows for ATP synthesis/enzymes in Calvin cycle allows

plant cells to use light energy in photosynthesis/DNA polymerase in DNA replication/RNA polymerase in transcription for genetic message to be read;;

28. Proteins like channel proteins/carrier proteins found on cell surface membrane embedded within the phospholipid bilayer to provide selective permeability;
29. E.g. (quote 1) Allow important biomolecules like glucose to enter for energy/allow for generation of membrane potential across neurones by regulating movement of ions
30. Proteins also found on cell surface membrane as receptors to allow communication between cells/with environment;;
31. E.g. GPRCR/Receptor tyrosine kinase;;

Accept AVP

AVP with quoted example

Argument again statement: carbohydrates are the most important component of living cells. (max 3)

Carbohydrates like glucose are important as the main respiratory substrate for production of ATP for cellular processs, used before lipids and proteins;;

Starch/glycogen serves as energy stores to store excess glucose;;

Cellulose/chitin play important role in plant cells to provide physical protection for plant cell/fungi;;

AVP

4 **The ongoing Covid-19 pandemic has resulted in worldwide disruption and millions of fatalities.** [25]

Discuss the underlying causes of pandemics and the difficulties in preventing future pandemics.

Related Learning Outcomes

Explain the factors affecting the probability that a pandemic will occur, including sanitation, water supply, food, climate, large-scale movements of people, evolution of new strains of virulent pathogens and development of drug resistance.

In order to score well, candidates should focus on:

- Biological factors as the primary cause of a pandemic
- Relevant trends affecting the probability of a pandemic
- The difficulties in implementing prevention methods addressing these

Candidates should avoid lengthy elaborations on the human immune response and should include the factors leading to a pandemic as underlying causes instead of being limited to only the cause of disease outbreaks.

Introduction:

1. Definition of a pandemic: An infectious fatal disease that has spread across multiple continents worldwide and affect a substantial number of people ;;

Format: Causes + Spread → Difficulty of prevention

Evolution -

1. Pathogens: Higher mutation rate of pathogen due to short generational time, lack of DNA repair mechanisms (or ssRNA being prone to mutations) leading to new variants;;
2. Evolution of new strains of virulence - Antigenic drift due to rapid transmission of pathogens and inherently high rates of mutations lead to higher chances of harmful variants developed ;;
3. Possible horizontal transmission of genetic material between pathogens lead to selection of highly resistant pathogen;;

Possible prevention (max 1)

4. Continuous research and development of antibiotics and careful administration of antibiotics ;;

   OR

   Implementation of vaccination programmes, with booster shots at regular intervals ;;

   OR

   Increased sanitation of hospitals and isolation of patients of infectious diseases to prevent horizontal transmission of antibiotic resistance genes / plasmids ;;

Difficulties

5. Inevitable continuous antagonistic co-evolution between pathogens and human antibiotic development leads to ineffective long term vaccination / necessity of booster shots;;
6. Rise of anti-vaccination sentiment / inability of certain groups to get vaccinated detracts from vaccination attempts / prevents herd immunity ;;
7. Long term indiscriminate usage of antibiotics lead to selection of drug resistant pathogen;;

Human-animal interactions –

8. High human-animal interaction through high rates of domestication give higher chances of increasing zoonotic transmission where a virus undergoes an antigenic shift, for eg. H1N1 virus;;
9. High human-animal interaction through hunting and consuming wild species and wild animal trade or other wildlife contact, eg. Ebola, Covid-19;;
10. Zoonotic pathogen spillover events increase pandemic risk by allowing virus to adapt better to spreading within human population, eg: coronavirus family;;

Prevention (max 1):

11. Having sanitary practices and quick containment procedures for livestock / Mandatory livestock and pet vaccination / Banning sale and consumption of bushmeat / Routine checks on and regulation of meat products ;;

Difficulty:

12. Difficulty of enforcing / policing continuous interactions between humans and animals as these interactions are necessary for livelihood ;;
13. Trend of increasing meat consumption due to various factors ;;

Climate change (AVP for 2021) -

14. Increased prevalence of disease vectors like parasites and insect vectors due to one more breeding cycle due to warmer winters and longer summers ;;
15. Disease vectors are able to increase their range from tropical to temperate zones due to increased global temperatures ;;

Prevention (max 1)

16. Keeping disease vector populations down by genetic modification, eg: Sterile male mosquitoes to bring down mosquito populations;;

    OR

    Preventing viruses from using insects as disease vectors, eg: genetic modification of mosquito to resist infection from dengue ;;

Difficulties

17. Measures taken need to be consistent to keep insect population down as resurgence in insect population is rapid due to low generational time and high offspring numbers ;;
18. Potential complications in reduction of insect vector species on the ecosystem ;;

High human population density –

19. Higher chances of transmission between humans through direct or indirect contact due to coming in contact with infected individual or breathing in droplets from sneezing or coughing, eg. Influenza/covid-19;;
20. High density of human population in cities give rise to potential overcrowding and increased challenges in sanitation and waste management, leading to fecal-oral infection from cholera/giardiasis/rotavirus/crytosporidiosis;;
21. Higher chances of interactions between humans and disease vectors from pests, leading to greater chances of disease outbreak, eg: Lassa fever due to Arenaviridae / Hantavirus pulmonary syndrome due to Bunyaviridae virus ;;

Prevention (max 1):

22. Preventing the rapid spread of pathogens through responsible containment methods – self-isolation, masks, Personal Protection Equipment ;;
23. Increased sanitation to prevent contaminated drinking water OR Urban waste management – preventing proliferation of disease vectors through excessive food wastage or preventing infection of fecal bacteria ;;
24. Keep the population of pests down by increased sanitation and food wastage management OR periodic extermination of pests ;;
25. Spreading out densely populated cities through technology and efficient public transport ;;

Difficulty:

26. Measures implemented require financial resources – potential segregation according to income OR disease outbreaks more prevalent in developing countries or regions ;;
27. Resistance or non-compliance with public measures leading to difficulty in implementing, policing and enforcement ;;
28. Physical limitations e.g. countries with limited land use for residential, economic activity
29. Necessity of interaction in some cases eg frontline/service workers OR in spaces eg for travel in bus/trains with restricted airflow

Movement –

30. Ease and convenience of traveling between countries and continents leads to increased likelihood of infections crossing continents;;
31. Rural-urban migration leading to overcrowding in cities;;
32. Migration patterns of animals traversing continents and countries leads to a possible spread of pandemic risk;;
33. AVP: War / conflict leading to refugees streaming into Europe and potentially a breakdown in public health facilities;;

Prevention (max 1)

34. Controlled travel for only vaccinated people or testing done prior to traveling / Minimizing long distance face to face meetings through using technology like zoom / Education for Personal hygiene – mask wearing

Difficulty

35. Long latency period of viruses lead to false negative tests of infected traveling individuals ;;
36. Groups of people missing out due to poor grasp / limited access to technology – elderly or low SES;;
37. Preventing of typical animal migration patterns would result in loss of species diversity / detrimental to species and ecological survival ;;